Wind Farm Layout Optimization

Using Aeroacoustic Surrogate Modeling and OpenFAST Validation

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Abstract

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Contents

1 Introduction

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1.1 Motivation

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1.2 Research Objective

The primary objective of this work is to investigate whether a computationally efficient surrogate-based optimization framework can be used to optimize wind farm layouts subject to both power production and aeroacoustic constraints.

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2 Background

2.1 Wind Farm Layout Optimization

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2.3 OpenFAST and FAST.Farm

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2.4 Aeroacoustic Modeling

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2.5 Genetic Algorithms

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3 Methodology

3.1 Overall Workflow

The optimization framework developed in this study consists of two key components: FLORIS and a custom machine-learning predictor for the steady-state aeroacoustic output of each turbine at a given receiver location. These computationally efficient approximators avoid the time and computational costs associated with repeatedly running OpenFAST, making them suitable for use within the genetic algorithm. However, they are not intended to serve as a complete replacement for a higher-fidelity simulation. For this reason, a final OpenFAST/FAST.Farm validation step is included.

During the optimization stage, a genetic algorithm evaluates candidate turbine layouts using FLORIS and the machine-learning aeroacoustic surrogate model. On the first iteration, candidates, also referred to as chromosomes, are generated from a uniform random distribution within the constraints of the land plot. FLORIS is used to estimate wake interactions, effective turbine wind speeds at the hub, and overall farm power production for each candidate. The effective turbine wind speeds are then supplied to the aeroacoustic surrogate along with the candidate layouts to estimate the A-weighted sound level at each receiver per turbine per candidate.

The outputs of these models are incorporated into the genetic algorithm by providing information to calculate candidate fitness. The fitness function takes into account power production, as well as turbine-spacing, farm-boundary, and aeroacoustic constraints to output a single float metric. To evaluate candidates under a range of operating conditions, each generation is evaluated across a set of weighted wind-speed and wind-direction scenarios.

After each generation, chromosomes are selected based on fitness and used to produce the subsequent generation through crossover and random mutation. A portion of the lowest-performing population is replaced with newly generated random candidates to maintain population diversity, while the highest-performing candidate is retained through elitism. The specific genetic operators and optimization parameters are described in greater detail in the following sections.

Once the genetic algorithm reaches its termination criterion, currently defined as a fixed number of generations, the highest-performing candidate is evaluated using OpenFAST/FAST.Farm. The resulting higher-fidelity aeroacoustic predictions are then used to recalculate the candidate's acoustic performance and fitness and to evaluate the agreement between the surrogate-based optimization and the OpenFAST validation.

3.2 Aeroacoustic Surrogate Model

Repeated OpenFAST AeroAcoustics simulations are computationally impractical within the genetic algorithm, where many candidate layouts must be evaluated across multiple wind conditions during each generation. A machine-learning surrogate was therefore developed to approximate the steady-state A-weighted sound level produced by an individual turbine at an arbitrary receiver location.

The surrogate was developed from single-turbine OpenFAST AeroAcoustics simulations spanning a range of wind speeds and observer locations. Multiple spatial feature representations and regression algorithms were evaluated to determine the combination that most accurately reproduced the OpenFAST results. The selected model was subsequently vectorized to efficiently predict the individual turbine contributions across all turbines, receivers, and candidate layouts required during optimization.

The following sections describe the initial stages of the basic interpolator, generation of the training dataset, the feature and model selection procedure, the final surrogate implementation, and the method used to aggregate individual turbine predictions into farm-level receiver sound levels.

3.2.1 Preliminary Surrogate Designs

Before developing the machine-learning surrogate, two simpler aeroacoustic prediction approaches were investigated. The first used a simplified analytical relationship between turbine sound-power level, wind speed, and propagation distance. The second incorporated OpenFAST AeroAcoustics results through interpolation over wind speed and observer azimuth. These preliminary approaches helped identify the variables and physical behavior that ultimately needed to be represented directly in the machine-learning surrogate.

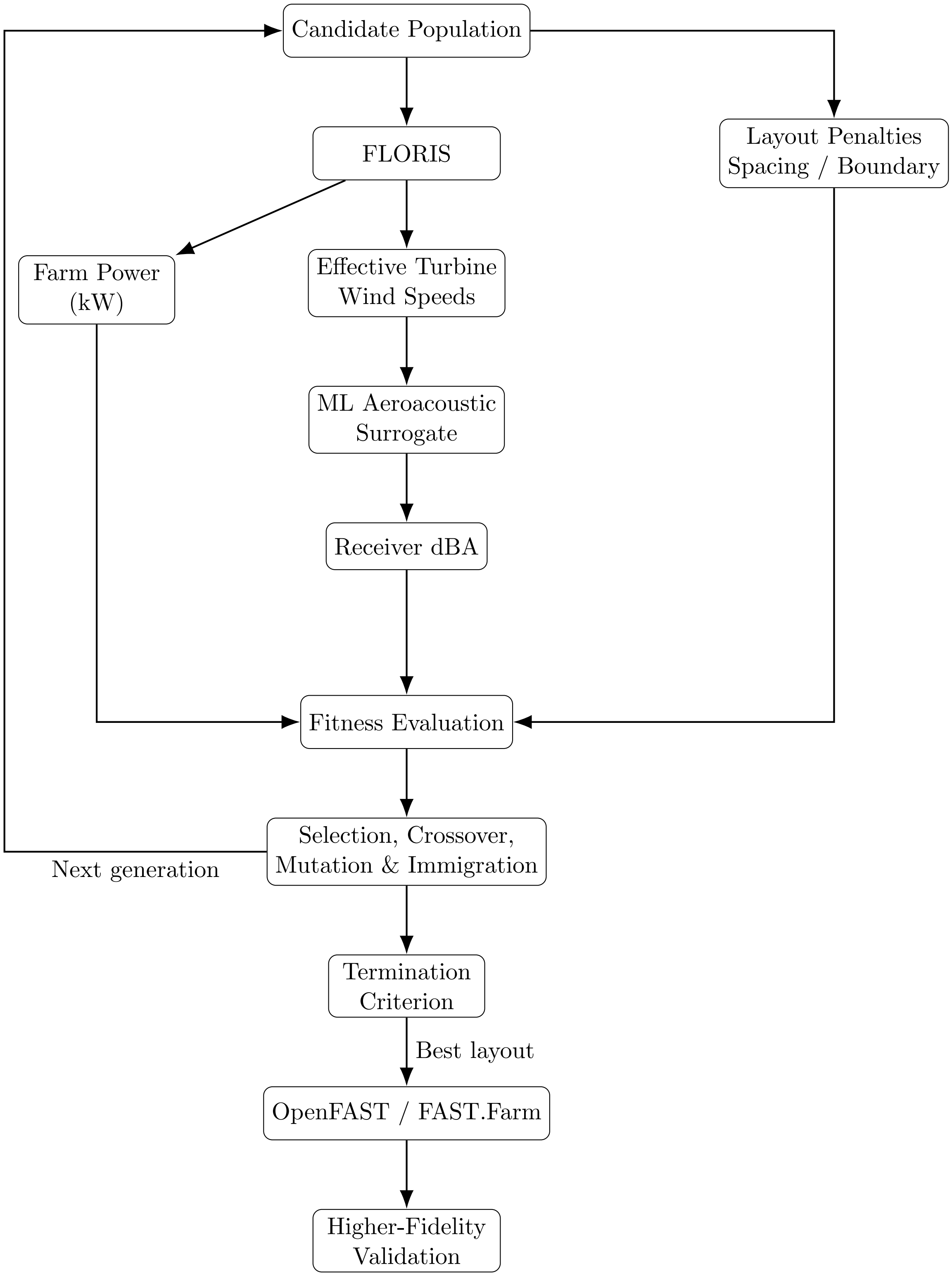


Figure 1: Overall genetic-algorithm optimization and OpenFAST/FAST.Farm validation workflow.

Simplified Analytical Model. The initial model estimated the A-weighted sound-power level of an individual NREL 5 MW turbine as a function of incoming wind speed. A piecewise relationship was assumed in which the turbine produced no acoustic contribution below 3 m/s, increased linearly from 90 dBA to 106 dBA between 3 and 11 m/s, and remained at 106 dBA for wind speeds greater than or equal to 11 m/s. The assumed relationship was

$$L_w(v) = \begin{cases} 0, & v < 3, \\ 90 + 2(v-3), & 3 \le v < 11, \\ 106, & v \ge 11, \end{cases} \tag{1}$$

where v is the incoming wind speed in m/s and L_w is the estimated A-weighted sound-power level.

The sound-pressure level at each receiver was subsequently approximated using

$$L_p(v,d) = L_w(v) - 20\log_{10}(d) - 8 - 0.005d, \tag{2}$$

where d is the three-dimensional turbine-to-receiver distance. This relationship was adopted as a simplified propagation approximation during early development and should not be interpreted as a complete implementation of an established acoustic propagation standard.

Although computationally inexpensive, this formulation depends only on wind speed and turbine-to-receiver distance. It therefore predicts the same sound level for receivers at equal distances regardless of their azimuth relative to the turbine and incoming wind direction. Examination of OpenFAST aeroacoustics results showed substantial directional variation, indicating that observer azimuth needed to be represented explicitly.

OpenFAST-Based Angular Interpolator. To incorporate this directional dependence, a second surrogate was constructed from single-turbine OpenFAST aeroacoustics simulations. Wind speed was varied from 3.0 to 24.5 m/s in increments of 0.5 m/s,

$$v \in \{3.0, 3.5, 4.0, \ldots, 24.5\}, \tag{3}$$

resulting in 44 distinct wind-speed conditions. For each condition, 32 acoustic observers were uniformly distributed around the turbine at a radius of 200 m.

The latter half of each aeroacoustics output time series was retained to approximate steady-state operation. An arithmetic mean of the A-weighted sound levels over this interval was then calculated for each observer. The resulting values were stored in `noise_lookup_table.json`, forming a two-dimensional lookup table parameterized by wind speed v and observer azimuth θ.

A coordinate transformation was required when coupling this dataset with FLORIS because FLORIS specifies wind direction using the meteorological "direction from which the wind originates" convention, whereas the aeroacoustic lookup requires the corresponding Cartesian source-relative direction. The conversion used in the implementation was

$$\theta_{\text{src}} = (90^\circ - \phi_{\text{met}})\,\frac{\pi}{180^\circ}, \tag{4}$$

where ϕ_{met} is the meteorological wind direction and θ_{src} is the corresponding Cartesian angle in radians.

The resulting OpenFAST data were interpolated over wind speed and azimuth using a linear `RegularGridInterpolator`. For receiver locations away from the 200 m reference radius, distance dependence was still approximated analytically using logarithmic geometric spreading and the constant 0.005 dBA/m attenuation coefficient.

This approach captured the strong directional variation absent from the initial analytical model, but the remaining analytical distance correction introduced another limitation. Comparison with additional OpenFAST simulations at varying observer distances showed that a single constant attenuation coefficient did not reproduce the distance-dependent behavior consistently across all observer azimuths and wind speeds. In some conditions the correction overestimated attenuation, while in others it underestimated it.

These observations motivated the final data-driven approach. Rather than interpolating only over wind speed and azimuth while prescribing the distance dependence analytically, the subsequent surrogate was trained to learn the joint relationship between wind speed, observer distance, observer direction, and A-weighted sound level directly from OpenFAST aeroacoustics data. A larger, homogeneous dataset was therefore generated for machine-learning model development, as described in the following sections.

It should be noted that this preliminary implementation averaged decibel values directly in the logarithmic domain. The later OpenFAST validation methodology instead uses energetic averaging, as described in Section 3.2.6.

3.2.2 Aeroacoustic Training Data

The training data for the aeroacoustic surrogate were generated using single-turbine OpenFAST simulations. Because each turbine was simulated independently, wake interactions between turbines were not included in the training-data generation process. Wake effects are instead accounted for later in the optimization workflow using FLORIS, which provides the effective wind speed experienced by each turbine.

Two primary variables were varied when generating the aeroacoustic training data: turbine wind speed and the position of the acoustic observer relative to the turbine. The wind-speed range was the same as Set (3).

Observer locations were defined in polar coordinates using radial distance r and azimuth angle θ. Twenty radial distances were generated using logarithmic spacing between 100 and 1000 m. The reference radius of 200 m was additionally included from the previous surrogate iterations to provide a higher-resolution angular sampling at a representative observer distance. Each of the logarithmically spaced radii was evaluated at 16 azimuth angles, while the 200 m reference radius was already evaluated at 32 azimuth angles.

The resulting observer sampling therefore contained

$$(20 \times 16) + 32 = 352 \tag{5}$$

observer positions. Combining these positions with the 44 wind-speed conditions produced a total of

$$N_{\text{samples}} = 44 \times 352 = 15{,}488 \tag{6}$$

single-turbine aeroacoustic training samples.

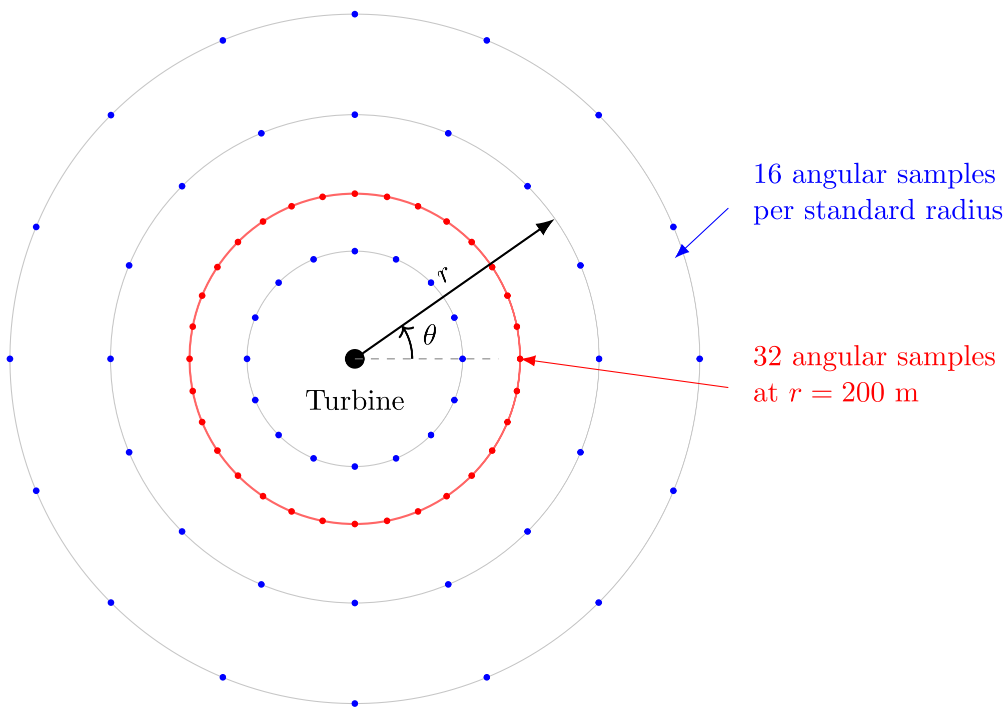
Table 1 summarizes the sampling strategy.

3.2.3 Candidate Feature Representations

The data generated from the OpenFAST simulations initially contained three input variables: wind speed v, relative observer coordinate X, and relative observer coordinate Y. Although the Cartesian

Table 1: OpenFAST aeroacoustic training-data sampling.

Variable	Sampling	Number of Values
Wind speed v	3.0–24.5 m/s, 0.5 m/s increments	44
Radial distance r	Logarithmically spaced, 100–1000 m	20
Azimuth θ	16 angles per logarithmically spaced radius	16
Reference radius	200 m with 32 azimuth angles	32 observer positions
Total observer positions	$(20 \times 16) + 32$	352
Total training samples	44×352	15,488



Schematic only: radial spacing is logarithmic from 100–1000 m

Figure 2: Schematic representation of the observer sampling used to generate the single-turbine aeroacoustic training dataset. Standard radial distances use 16 uniformly distributed azimuth angles, while the 200 m reference radius uses a denser 32-angle sampling.

coordinates fully describe the observer location, the aeroacoustic field exhibits a strong radial and directional structure. Several alternative feature representations were therefore evaluated to make this underlying geometry more explicit to the machine-learning models.

The relative Cartesian coordinates were first transformed into polar coordinates according to

$$r = \sqrt{X^2 + Y^2}, \tag{7}$$

and

$$\theta = \operatorname{atan2}(Y, X). \tag{8}$$

Directly using θ as a numerical feature introduces an artificial discontinuity at the angular boundary. For example, directions immediately below 2π and immediately above 0 are physically adjacent but numerically distant. To preserve the periodic nature of the observer azimuth, the angular coordinate was therefore encoded using

$$c_\theta = \cos(\theta), \qquad s_\theta = \sin(\theta). \tag{9}$$

Distance was also represented using several transformations. Under geometric spreading, acoustic intensity decreases with increasing distance, while sound-pressure level is expressed on a logarithmic scale. A logarithmic transformation of the observer distance,

$$r_{\log} = \log_{10}(r), \tag{10}$$

was therefore evaluated as a candidate feature. Reciprocal distance terms were additionally considered,

$$r^{-1} = \frac{1}{r}, \qquad r^{-2} = \frac{1}{r^2}, \tag{11}$$

to determine whether explicitly representing distance-decay relationships improved predictive performance.

Rather than selecting one representation a priori, five candidate feature sets were constructed and evaluated with each candidate machine-learning model. These feature sets are summarized in Table 2. The complete set of original and engineered features is retained in `training_data.csv`.

Table 2: Candidate feature representations evaluated for the aeroacoustic surrogate.

Feature Set	**Input Features**	**Purpose**
Cartesian	$v,\ X,\ Y$	Original spatial representation
Polar	$v,\ r,\ \cos\theta,\ \sin\theta$	Explicit radial and directional representation
Polar + log distance	$v,\ \log_{10}(r),\ \cos\theta,\ \sin\theta$	Logarithmic distance representation
Polar + raw/log distance	$v,\ r,\ \log_{10}(r),\ \cos\theta,\ \sin\theta$	Combined raw and logarithmic distance
Polar + distance decay	$v,\ \log_{10}(r),\ r^{-1},\ r^{-2},\ \cos\theta,\ \sin\theta$	Multiple representations of distance decay

The feature representation was not selected independently of the regression model. Instead, each candidate machine-learning algorithm was trained and evaluated using all five feature sets. The final combination of model architecture and feature representation was selected based on predictive performance, as described in the following section.

3.2.4 Model Training and Selection

Seven regression algorithms were evaluated for the aeroacoustic surrogate: Linear Regression, Ridge Regression, Polynomial Ridge Regression, Random Forest, Extra Trees, Histogram Gradient Boosting, and eXtreme Gradient Boosting (XGBoost). These models were selected to provide a range of model complexities, from simple linear baselines to nonlinear ensemble and gradient-boosting methods.

Table 3 summarizes the candidate regression algorithms.

Table 3: Candidate regression algorithms evaluated for the aeroacoustic surrogate.

Model	**Model Type**	**Role in Comparison**
Linear Regression	Linear	Simple baseline model.
Ridge Regression	Regularized linear	Linear model with L_2 regularization applied.
Polynomial Ridge	Nonlinear polynomial	Higher-degree polynomial model with L_2 to avoid overfitting.
Random Forest	Tree ensemble	Nonlinear ensemble based on averaging predictions from multiple decision trees.
Extra Trees	Tree ensemble	Highly randomized tree ensemble providing an alternative to Random Forest.
Histogram Gradient Boosting	Gradient boosting	Nonlinear boosted-tree model designed for efficient regression on continuous data.
XGBoost	Gradient boosting	Alternative gradient-boosted tree implementation used to compare performance with Histogram Gradient Boosting.

Model architecture and feature representation were evaluated jointly. Each of the seven regression algorithms was trained using each of the five feature representations described in Table 2, resulting in

$$N_{\text{configurations}} = 7 \times 5 = 35 \tag{12}$$

candidate model configurations.

The dataset was divided into training and testing subsets using an 80/20 split. The same split was used for all model and feature-set combinations to provide a consistent basis for comparison.

Predictive performance was evaluated primarily using mean absolute error (MAE) and root mean squared error (RMSE), between the A-weighted sound level obtained from OpenFAST aeroacoustics, L_i, and the corresponding surrogate prediction $\hat{L}_i$. The coefficient of determination, R^2, was also calculated as an additional measure of model fit.

The final surrogate configuration was selected based on its performance on the held-out test data. The comparative results of the 35 candidate configurations are presented in Section 4.1.

3.2.5 Final Surrogate Implementation

The machine-learning surrogate uses the input feature vector

$$\mathbf{x} = \left[v, \quad \log_{10}(r), \quad \cos(\theta), \quad \sin(\theta)\right]. \tag{13}$$

The surrogate then approximates the relationship

$$\hat{L}_A = f\left(v, \log_{10}(r), \cos(\theta), \sin(\theta)\right), \tag{14}$$

where $\hat{L}_A$ is the predicted A-weighted sound level.

3.2.6 Aeroacoustic Aggregation

Independent turbine noise contributions are combined in the linear acoustic-energy domain:

$$L_{\text{total}} = 10 \log_{10}\left(\sum_{i=1}^{N} 10^{L_i/10}\right). \tag{15}$$

Time-varying sound levels may similarly be reduced using

$$L_{\text{eq}} = 10 \log_{10}\left(\frac{1}{T}\sum_{t=1}^{T} 10^{L_t/10}\right). \tag{16}$$

3.3 Genetic Algorithm

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3.3.1 Chromosome Representation and Initialization

3.3.2 Fitness Function

The optimization objective can be expressed generally as

$$F = P_{\text{expected}} - C_{\text{spacing}} - C_{\text{noise}} - C_{\text{boundary}}, \tag{17}$$

where P_{expected} represents expected farm power and the remaining terms represent constraint penalties.

3.3.3 Wind Scenarios

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3.3.4 Genetic Operators

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3.4 OpenFAST Validation

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4 Results

4.1 Surrogate Model Performance

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An example results table is shown in Table 4.

Table 4: Example surrogate model performance.

Model	MAE (dBA)	RMSE (dBA)	R^2
Model A	0.10	0.15	0.999
Model B	0.15	0.22	0.997
Model C	0.20	0.28	0.995

4.2 Genetic Algorithm Performance

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4.3 Optimized Layout

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4.4 OpenFAST Validation

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5 Discussion

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6 Limitations

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Example limitations could be presented as:

- OpenFAST validation currently uses deterministic uniform inflow.
- The aeroacoustic surrogate is valid only within the domain represented by its training data.
- The Genetic Algorithm does not guarantee identification of the global optimum.
- Additional turbulence conditions would require further high-fidelity simulations.

7 Future Work

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8 Conclusion

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References

[1] Author, *Example OpenFAST Reference*, Publisher or Journal, Year.

[2] Author, *Example FLORIS Reference*, Publisher or Journal, Year.

A Optimization Parameters

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B Additional Results

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